



Data Engineering Fundamentals Assignment

By Modinat Titilayo Ganiyu

BUSINESS SCENARIO

Beejan Technologies is a telecom company

Everyday, thousands of customers complain about issues like poor network, incorrect billing or bad customer service.

The management is frustrated. Data is stored in different format. The reporting team manually compiles spreadsheet. No single pipeline exists for this data flow. Report are delayed. Teams work in silos.



Beejan

Technologies

Connecting people, everywhere

Current Challenges



Beejan Technologies currently faces several challenges

- Teams work independently, making it difficult to share and integrate data.
- Scattered complaint data across social media, SMS, call center logs, and website forms.
- The reporting team manually collects and combines data from different sources.
- Inconsistent data formats.
- Delayed reports
- Management does not have a single, timely view of customer complaints and emerging issues.



PROPOSED SOLUTION

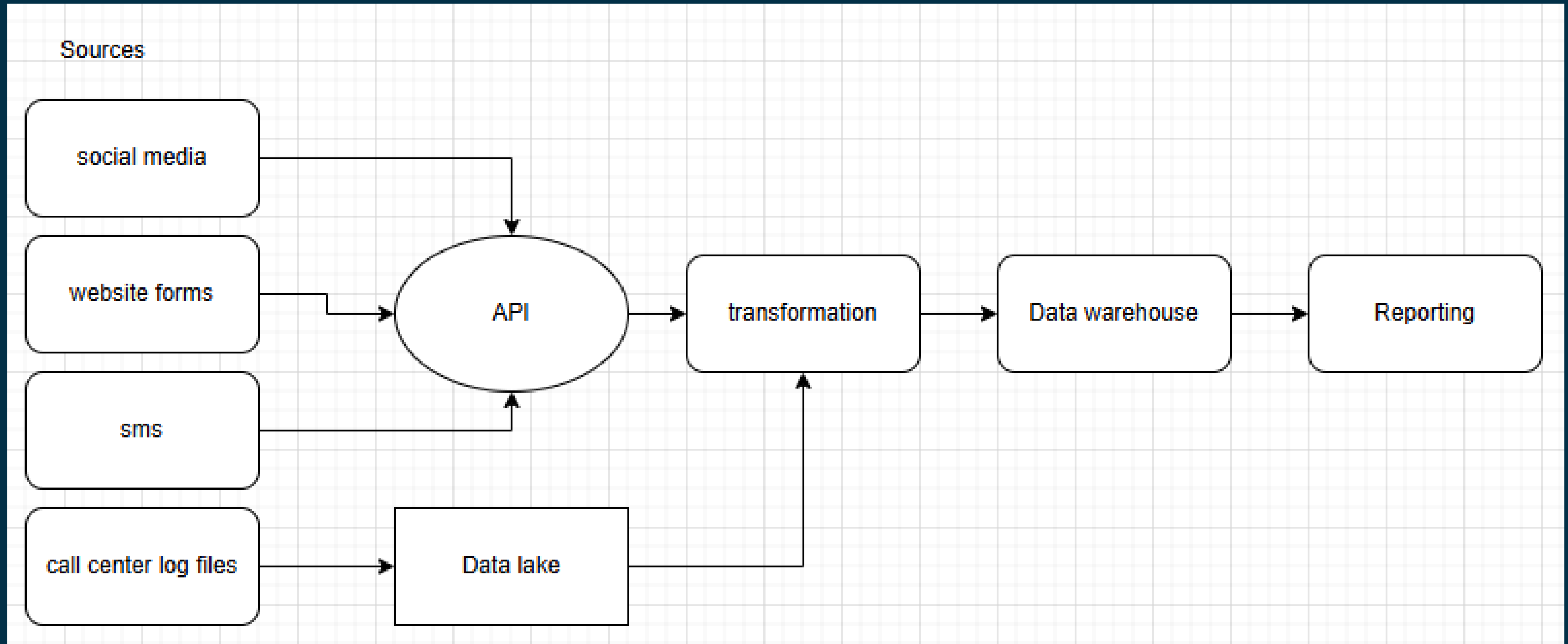
Proposed Data Engineering Solution



I will build an automated end-to-end data pipeline that collects complaint data from all four sources, processes and standardizes it, stores it centrally, and makes it available for reporting and decision-making.

Collection → Ingestion → Transformation → Storage → Serving → Report

Architecture Flow Diagram



MEET THE TEAMS



Marketing Team

This team provides complaints from all social media platform.



IT team

This team provides complaints through website forms.



Customer Service Representatives

This team provides complaints from call center log files and sms.



DATA COLLECTION

Data Contract



I'll work closely with data collection teams responsible for website forms and call center complaints to establish data contract . This ensures the data collected can be useful, consistent, trustworthy and sustainable in the long term.

The data contract will clearly define:

- what to capture
- what values are allowed
- what format to use
- what to do when information isn't available

The data contract becomes a standard guide for each team, ensuring that customers complaint are consistently captured as structured records before they enter the data system.

IT Team — Website Form



The website form should enforce the required fields and predefined options at the point of submission, ensuring that complaints are captured consistently and correctly before entering the ingestion system..

field	requirement	Format
customer_name	Customer's full name	Text
phone_number	Customer’s phone number	+23412345678
complaint_category	Must be selected from: Poor Network, Incorrect Billing, Bad Customer Service	Dropdown
complaint	Original complaint from the customer	Text
complaint_timestamp	Date and time the issue occurred	YYYY-MM-DD HH:MM

Customer Service Team – Call Center

The call center system should enforce these fields and predefined values when agents record complaints, ensuring consistent and structured data before ingestion.



field	requirement	Format
customer_name	Customer's full name	Text
phone_number	Customer's phone number	+23412345678
complaint_category	Must be selected from these 3 categories: Poor Network, Incorrect Billing, Bad Customer Service	Predefined value
complaint	Complaint recorded during the call	Text
call_id	Unique ID for each call	CALL00123
complaint_timestamp	Date and time the complaint was made	YYYY-MM-DD HH:MM

DATA SOURCE 1:

SOCIAL MEDIA



Customers post complaints on social media such as:

“Your network has been terrible since morning. I can’t make calls or browse.”

The social media platform's API will be used to continuously collect new comments on Beejan Technologies social media platform.

DATA SOURCE 2:

WEBSITE FORMS

—————>

- customer name: Ola Ahmed
- phone_number: +2348012345678
- complaint_category:
[Incorrect Billing ^]
- complaint: I was debited twice for the same data plan.
- complaint_timestamp: YYYY-MM-DD HH:MM
2026-06-23 08:23

customer name	phone_number	complaint_category	compaint	complaint_datetimestam p
Ola Ahmed	+2348012345678	Incorrect Billin	I was debited twice for the same data plan.	2026-06-23 08:23

Here’s how IT team is going to structure the complaint.
The structure is enforced when inputting the answers for each field

This data will be collected and stored in database table

→

DATA SOURCE 3:CALL CENTER LOG FILES

Here’s how the customer service representative team is going to structure the complaint received through phone call from customers.

- Following the data contract
- Following the fomate
- Strictly entering values that are allowed

call_id	customer name	phone_number	complaint_category	compaint	complaint_datetim estamp
CALLO0123	Ali	0812345678	Bad Customer Service	Customer is unhappy because their previous complaint was not resolved.	2026-05-21 18:15

DATA SOURCE 4:

SMS



Customers sends complaints through SMS such as:

“I was charged twice for my subscription.”

An API will be used to receive each SMS complaint as soon as it arrives, then pass it into the streaming pipeline



INGESTION

Ingestion Strategy

- Incremental loading will be used so that only new complaints are loaded instead of reloading the entire dataset everytime. This reduces processing time and prevents unnecessary storage growth.
- Call center log files will be stored as Parquet files in the data lake and ingested through a file-based batch process.
- API ingestion will be used to load complaints from social media, SMS, and website forms into the streaming pipeline.

Ingestion Frequency



A hybrid approach of streaming and batch ingestion will be used:

- Social Media → Streaming:

Complaints arrive continuously, so they will be ingested in real time which will allow the team to detect issues quickly and respond as soon as complaints are received.

- SMS → Streaming:

Customers can send complaints at any time. And the team should be able to respond to it without waiting. So, streaming allows each SMS complaint to be captured and ingested immediately after it is received.

- Website Forms → Streaming:

Each form submission can be sent to the API and process immediately.

- Call Center Logs → Batch:

Call center complaints are recorded as log files, which can accumulate over a period of time. Instead of processing each call immediately, the files can be collected and processed in batch (e.g., every hour).



TRANSFORMATION

UNSTRUCTURED DATA



Since complaint from sources such as SMS and social media are unstructured, an NLP/ML classification model will be used during transformation to automatically classify each complaint into one of the three categories: Poor Network, Incorrect Billing, or Bad Customer Service.

For example:

complaint: "My data connection has been terrible since morning."
→ Poor Network

complaint: "I was charged twice for my subscription."
→ Incorrect Billing

complaint: "The customer service agent was rude to me."
→ Bad Customer Service

CLEAN AND STANDARDIZE



- Standardize column names across all sources.
- Fix inconsistencies
- Check and remove duplicates.
- Handle missing values.
- Validate required fields and data types.
- Verify that each captured change is valid, accurate, and correctly applied to the target system.



STORAGE

STORAGE STRATEGY



- Call Center Logs → Data Lake (Parquet):

Call center log files will first be stored in the data lake because they're recorded as log files and as parquet files because Parquet is columnar, compressed, machine-readable, and more storage-efficient for large volumes of log data.

- SMS, Social Media & Website Complaints → Data Warehouse:

Complaint data from all sources will be stored in the data warehouse as the centralized storage layer for standardized complaint data, regardless of whether the data is ingested through streaming or batch processing.



SERVING

REPORTING



The transformed data in the data warehouse will be queried using SQL to generate the required metrics and insights.

The results will be presented through a dashboard, making the reports easily accessible to downstream users for monitoring and decision-making.

The dashboard will show metrics such as:

- Top complaint issues
- Dates with the highest number of Poor Network complaints
- Channel with the highest number of complaints
- Complaint trends over time
- Complaint volume by category



ORCHESTRATION & MONITORING

ORCHESTRATION



The pipeline will use orchestration to schedule, coordinate, and monitor all tasks.

How often will it run?

Because the pipeline uses both streaming and batch:

- Social Media → Streaming: Runs continuously.
- SMS → Streaming: Runs continuously.
- Website → Streaming: Runs continuously.
- Call Center Logs → Batch: Runs on a schedule, e.g. every hour.

The orchestrator will monitor each pipeline task and detect failures.

When a task fails, alert is sent through email, Slack, Microsoft Teams, or other notification channels.



DATAOPS

PRODUCTION



The pipeline will run on a cloud server.

The pipeline will be made available in production by:

1. Packaging the pipeline with its dependencies.
2. Deploying it to a production cloud server environment.
3. Configuring production data sources, databases, storage, and credentials.
4. Scheduling and starting the pipeline so streaming processes run continuously and batch processes run at their defined schedule.
5. Monitoring the production pipeline and sending alerts when failures occur.

In short:

Develop → Test → Package → Deploy → Configure → Run → Monitor.



THANK YOU!